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Profile

Electronics Engineer with strong interdisciplinary expertise in Artificial Intelligence, Embedded Systems and Data Science. Skilled in developing intelligent systems combining hardware and software, including IoT solutions, machine learning models and real-time applications. Passionate about building scalable AI driven products and solving real world problems through innovation in signal processing, analog digital electronics and generative AI. Currently expanding expertise in PCB Design, RTL Design, and GDS flow using Verilog, along with system level design and hardware software co design.

Projects

1. AI SINGLE COPILOT SYSTEM

- Developed an AI-based decision support system for aviation assistance
- Integrated real-time data such as fuel, weather and aircraft parameters
- Implemented predictive models for alternate route and fuel optimization

2. SMART CAR PARKING SYSTEM

- Designed an IoT based parking management system
- Used sensors to detect slot availability in real time
- Improved parking efficiency and reduced congestion

3. Local Business Prediction AI System

- Developed AI chatbot to suggest profitable business ideas based on location
- Includes risk analysis, ROI prediction and demand mapping
- Integrated Google Maps and trend analysis concepts

4. Smart Dustbin (IoT-Based)

- Designed an automated waste management system
- Uses sensors for level detection and alerts
- Helps optimize waste collection and hygiene

5. Number Plate Detection System

- Built a computer vision model for vehicle number plate recognition
- Used image processing and ML techniques
- Applicable in surveillance and traffic management

6. SafeNest – Disaster-Resilient IoT Communication System

- Built SafeNest, an ESP32 and LoRa based emergency communication system.
- Enabled long range transmission of GPS and biological data without cellular or internet connectivity.
- Implemented reliable wireless communication using ACK based LoRa protocol.

Education-Bachelor's Degree in Electronics Engineering

RAJALAKSHMI INSTITUTE OF TECHNOLOGY, CHENNAI

Skills & Abilities

- | | |
|--|--|
| • Python | Analog Circuit Design |
| • C programming | • Digital Electronic and Signal Processing |
| • Data science and Machine Learning | • Embedded System & IoT |
| • Generative and Agentic Model (BERT, GPT AND LANGCHAIN, RAG MODE) | • Assembly Language |
| | • Web Technologies (Frontend -HTML, CSS, Backend-Django) |